

CLASS 05 / SEP 22 / U04

ARRAYS AND LISTS

PART TWO

STAGGER IT, SPIN IT, BEND IT

P1b DUE

U04 CONTINUES / P2a IN PROGRESS, DUE AT THE MIDTERM

FOUR ARRANGEMENTS

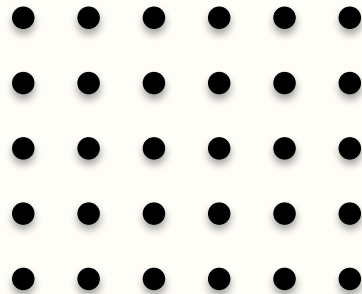
Same component, four rules for generating the list of positions.

LINEAR



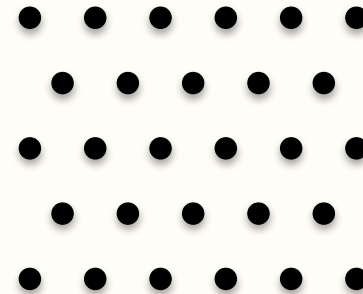
one counter

GRID



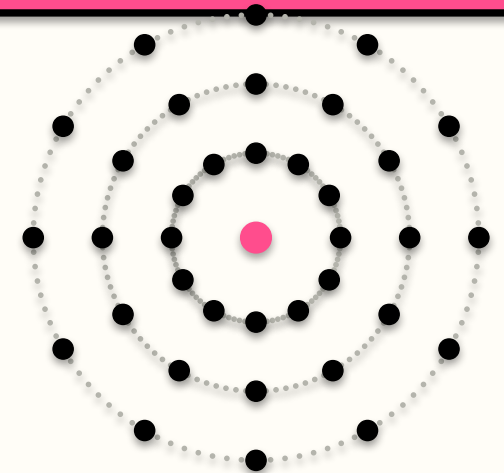
two counters

HEXAGONAL



stagger alternate rows

RADIAL



radius, count, angular step

Every one of these is the same operation — move, repeated, with the amount driven by a counter. Only the rule for the counter changes.

HEXAGONAL IS A GRID WITH AN ODD/EVEN TEST

Stagger alternate rows by half the spacing. Keep the row spacing consistent.

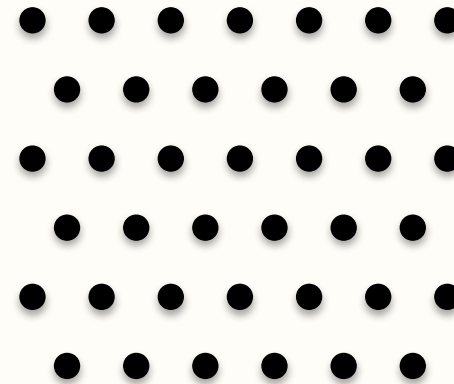
THE TEST

```
offset = (j mod 2 == 0) ? 0 : dx / 2
```

Modulo asks: is this row odd or even? Compare turns that into True or False, and Switch picks the offset. The same two nodes as last week.

Row spacing is not dx. If you use the same number for both, you get a squashed grid, not a hex field.

WHAT IT LOOKS LIKE



A BRICK BOND IS A HEXAGONAL ARRAY. SO IS A HONEYCOMB. SO IS HALF THE PAVING YOU WALK ON.

RADIAL IS POLAR ARITHMETIC

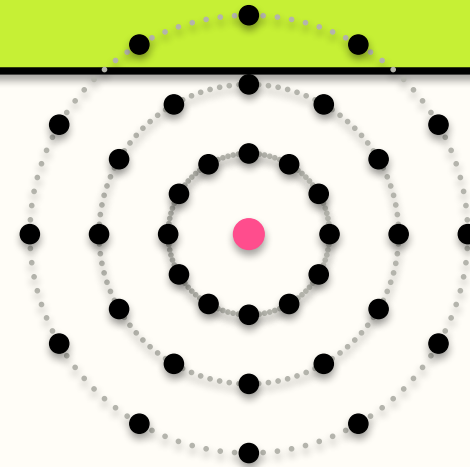
Radius, count, angular step. The index drives the angle.

THE RULE

```
step = 360 / count  
 $\theta[i] = i \times \text{step}$   
 $p[i] = c + r \cdot (\cos \theta, \sin \theta)$ 
```

The same rotation arithmetic from Week 2 — the one place where x and y have to talk to each other — now generating positions instead of moving a solid.

TWO KNOBS



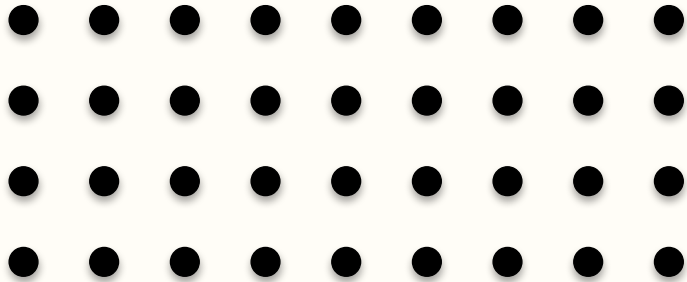
Change the count and the step changes with it. Change the radius per ring and it opens up.

IF YOUR LAST ELEMENT LANDS ON TOP OF YOUR FIRST, YOU DIVIDED BY THE WRONG NUMBER.

MOVE ONLY SOME OF THEM

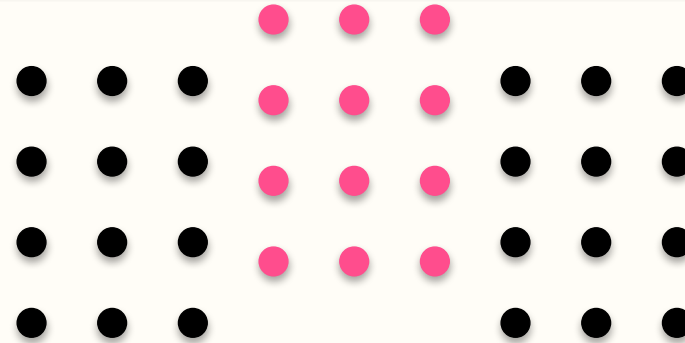
Select a range of indices with comparisons, and displace just those.

BEFORE



Every element identical. The index is computed, then ignored.

AFTER



Indices 3 to 5 lifted. One Compare, one Switch, one Set Position.

THE FIELD STOPS BEING WALLPAPER THE MOMENT A TEST DECIDES SOMETHING.

SINE GIVES YOU A SMOOTH RULE

Math set to Sine, driven by the index. Three words describe the whole curve.

THREE PARAMETERS

AMPLITUDE

how far it swings

FREQUENCY

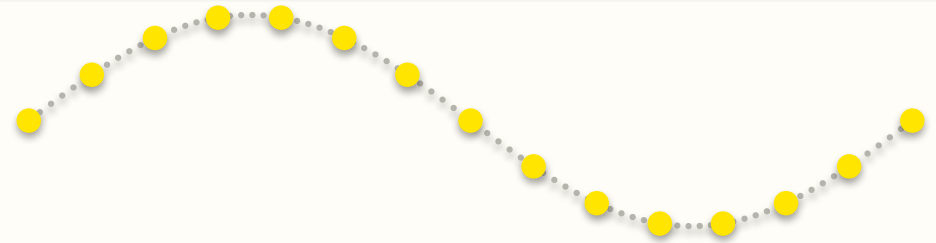
how many swings across the set

PHASE

where in the swing it starts

$$h[i] = \text{amp} \times \sin(i \times \text{freq} + \text{phase})$$

READ IT AS A CURVE



Connect the ordered points and the list becomes a curve. That is the same list — read as a shape instead of as positions.

NAME THE THREE VALUES ON YOUR SHEET. P2a ASKS FOR THEM EXPLICITLY.

TWO WAYS TO REPEAT

Worth being precise about, because they look alike and are not.

EXPLICIT ITERATION

A Repeat Zone runs a body once per step, in order, carrying a value forward.

Use it when step n depends on step $n-1$ — and when you want to watch the loop happen.

PER-ELEMENT EVALUATION

The other kind is evaluated across every element at once. There is no order and no carry.

It is faster and it is usually what you want — but nothing on screen tells you a loop happened, which is exactly why we built the explicit one first.

NEXT WEEK THIS GETS A NAME. FOR NOW: ONE CARRIES A VALUE FORWARD, THE OTHER DOES NOT.

THE SAME NINE LINES, A WEEK LATER

Last week you could read the first three lines. Today you can read all of it.

```
points = []

for j in range(rows):                # nested loop = grid
    offset = spacing / 2 if j % 2 else 0  # odd/even stagger
    for i in range(cols):
        z = amplitude * sin(i * frequency + phase)
        if 3 <= i <= 5:                # Compare + Switch
            z = z + lift
        points.append([i * spacing + offset, j * row_gap, z])
```

WHAT IT MAPS ONTO

list	modulo
index	sine
count	amplitude
loop	frequency
nested loop	phase
conditional	

j % 2 IS THE ODD/EVEN TEST · if IS COMPARE AND SWITCH · sin IS THE MATH NODE

NOW — FINISH THE FOUR

Rest of the session: hexagonal, radial, the conditional and the sine.

✓

Hexagonal array working, with the odd/even test visible and row spacing correct

✓

Radial array working from radius, count and angular step

✓

One index range displaced — and you can say which indices and why

✓

Sine driving height, with amplitude, frequency and phase written down

✓

Three of the four chosen for the sheet, including the nested grid

P2a IS THREE COMPARISONS, NOT FOUR UNRELATED PICTURES.

BEFORE NEXT CLASS

WATCH

U05 — Attractors. Distance, remap, clamp and falloff. Next week the rule stops coming from the index and starts coming from a measurement.

BRING

Your four arrangements, working. We drive them with attractors next class, so bring a file you can build on.

CHECK

P1c — print-ready geometry, physical units, orientation and the sliced preview get reviewed today before anything is queued.

FINISH

P1b is due today. P2a is due at the midterm; keep the comparisons moving.